Using Machine Learning To Improve The LUX-ZEPLIN Experiment

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Abstract

In this review, the evidence for and techniques used in searching for WIMP dark matter via scintillation from nuclear recoils are discussed. Detecting these recoils requires the passing of an emitted photon through a photomultiplier tube and the reconstruction of the properties of the event out of the characteristics of the resultant waveforms, but the stochastic traits of the method introduce error. A notable use case for this method is the LUX-ZEPLIN experiment, which is a liquid xenon-based direct detection experiment and the focus of this review. The existing structure and reconstruction methods of the LUX-ZEPLIN detector are reviewed to identify weaknesses in its current implementation. A neural network is then trialled for potential improvements in reconstruction performance compared to the conventional case.

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1 Introduction

One of the most recognised dark matter (DM) theories in the current day is that of the weakly interacting massive particle (WIMP), which posits that DM might be a particle from beyond the standard model that interacts only by gravity, and very occasionally through the weak force. To verify this theory, various approaches can be taken, one of which is to directly detect an interaction between a baryonic matter particle and a WIMP. One possible interaction channel is the interaction between a WIMP and a nucleus, which causes the nucleus to recoil. The LUX-ZEPLIN experiment (LZ) is one such experiment, which is a large tank placed deep underground to minimize background noise and filled with liquid xenon to use as a target. The xenon is chosen because it scintillates easily and suspended in a component called the "time projection chamber" which is built to detect interactions from multiple vectors. On interaction, the emitted photon is detected by photomultiplier tubes (PMTs), and an emitted electron drifts to the top of the chamber where it is extracted into a gaseous electroluminescent region. These signals together can be used to reconstruct the position and characteristics of the interaction. Despite being almost a mile underground, the LZ still experiences background radiation from things like neutrinos or contamination by radioactive isotopes in the detector's hardware, which is why reconstruction is so important to ensure that a detected signal is DM and not background. This is currently achieved by a software that integrates over the voltage distributions produced from particles detected by the PMTs for a net result, but this is a cumbersome process and not without error, which will be discussed later. However, in recent times machine learning (ML) models have proven their worth for mathematical and scientific purposes including pattern recognition and classification. As reconstruction is a mathematical task using principles of classification and pattern recognition, it stands to reason that a ML procedure could improve or replace the existing photon counting algorithm with better results, which is the main idea presented in this review.

2 A Primer On Dark Matter

2.1 History

In 1933, astronomer Fritz Zwicky observed that most galaxy-sized objects and above should not be able to gravitationally bind, owing to the high rotational velocities of their constituents. He chose to investigate this by using the virial theorem[1]:

$$2\langle K\rangle = -\langle U\rangle \tag{1}$$

$$\frac{5}{3}\frac{r}{G}\sigma_v^2 = M_{tot} \tag{2}$$

He observed the Coma Cluster's visible mass and rotational velocity to then get a value

for the virial mass, which he compared and found that the Coma Cluster's virial mass appeared to be a hundred times more than its visible mass. He coined the term "dark matter" to explain his hypothesis, which was that the vast majority of the mass in the Cluster was invisible and only contributed to the Cluster gravitationally.

As this claim was incredibly bold, it was of course not immediately accepted, but later work would cause the hypothesis to rise in prominence. In 1970, astronomers Albert Bosma, Morton Roberts, and Ken Freeman chose to observe other galaxies to make sure that this gravitational anomaly was widespread; they did this by using the laws of gravity to predict how their constituent objects should behave, then observing their actual behaviour. The observed rotation curves for the constituent objects of galaxies did not line up with the predicted curves, confirming that something did not align between the theory and observations of gravity.[2]

In response, multiple theories were presented. Dark matter was one of them, but another was the idea that the theory of gravity at galactic scales needed revising. Evidence to settle this debate would come when the Bullet Cluster was photographed by NASA, showing a galaxy merger in progress. A notable feature of the Cluster is that the center of gravitational lensing is offset relative to the center of visible mass, which is incompatible with simple theories of modified gravity that do not invoke a new scalar field or other similar concepts that could be considered "dark matter". This is because modified Newtonian dynamics is a modification of Newtonian gravity, which relies on the Standard Model as a source of gravity. However, this model does not predict a static invisible gravitational source, but the theory of dark matter does.[3]

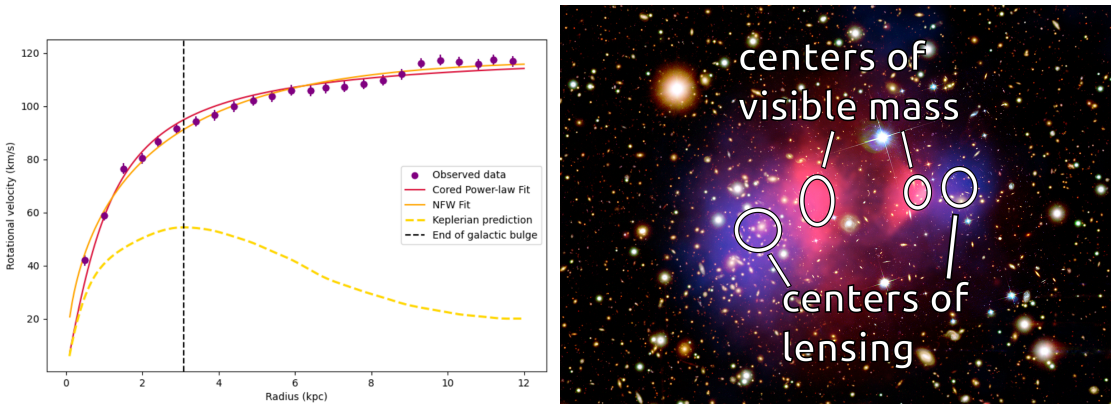


Figure 1: Left: the angular velocity of constituents at a certain distance from the center of the M33 galaxy. The distributions visibly do not agree with the Keplerian prediction. Made in matplotlib. Right: the Bullet Cluster, composite public domain image from ESA. Captioned by E. Ducluzeau to show centers of mass and lensing.[4]

Even more evidence for DM can be extracted from the cosmic microwave background (CMB), which has a fluctuating power spectrum. The CMB is the last photons scattered at recombination, which is the epoch where the primordial plasma of drifting subatomic

particles first combined into atoms and the universe transitioned from opaque to transparent, preventing large-scale photon scattering. Multiple models of recombination can be simulated, and the one that produces the CMB anisotropies that align with what current observations is the one which includes early universe DM as a source of gravitational anomalies that can cause fluctuations in the CMB's density, leading to a change in scattering profiles and the introduction such anisotropies.[5] Currently, the theory of DM is widely accepted as the source of these anomalies, leading to the question of what exactly it is.

2.2 Weakly Interacting Massive Particles

One theory of dark matter is that it is a particle. The first theories of particle DM inspected the Standard Model and particularly the neutrino, which is an invisible massive particle that always moves at ultrarelativistic speeds, but two conflicting bounds on the masses of neutrinos and DM exist which forbid this theory. In order for the observed neutrino masses to be correct, the upper bound on neutrino relic density must be:

$$\Omega_\nu h^2 = \frac{\sum m_\nu}{100\text{eV}} \tag{3}$$

but the Tremaine-Gunn bound of

$$M = m_f \times V \times \int f(\vec{p}) d^3p \tag{4}$$

set at $m_f = 120\text{eV}$ for a DM relic density of:

$$\Omega_c = 1 - \Omega_b - \Omega_k - \Omega_\Lambda \tag{5}$$

with the result $\Omega_c h^2 = 0.12$ gives the minimum mass of a DM candidate must have to fit the known DM density of galaxies. Since $\Omega_\nu h^2$ cannot exceed 1, the maximum mass of the neutrino must be 100 eV, but the Tremaine-Gunn bound sets a minimum DM mass of 120 eV; this contradiction means that the neutrino cannot be DM.[6] A search beyond the Standard Model is required for a particle that only interacts through gravity and very rarely by the weak force. This hypothetical particle is the "weakly interacting massive particle", or "WIMP" for short.

This prediction is motivated by the accidental prediction of a DM particle by the theory of supersymmetry (SUSY). SUSY was formulated to solve the problem of force reunification where the fundamental forces of physics' scales cross over at incredibly similar values but never quite unify. SUSY predicts that each Standard Model boson and fermion has a "superpartner" of the opposite type through an extension in the O(1) of the Poincare group:

$$m_{new} \sim m_{SM} + m_H \sim 100\text{GeV} \tag{6}$$

A group of particles predicted by SUSY is the "neutralinos", fermionic superpartners of the bosons, which would only interact gravitationally and very rarely via the weak but not travel at ultrarelativistic speeds, leading to the neutralino having higher possible masses.

Further investigation into the neutralinos' properties reveals that they have the exact properties required to act as a DM candidate obeying the Tremaine-Gunn bound and low enough velocities to cluster together and form the large scale DM structures required for the universe to have its current shape[7]. SUSY has not yielded results since it was theorised

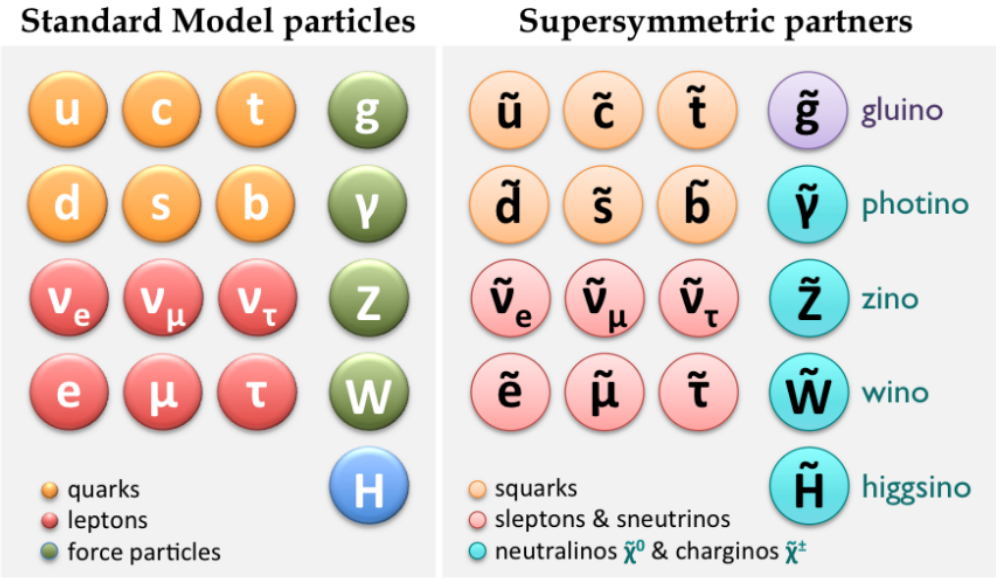


Figure 2: A table of predicted superpartners to the Standard Model particles. Note the presence of the fermionic neutralinos.[8]

and exploring it is getting more and more difficult, leading to a decline in SUSY research, but WIMP theory remains popular as a theory that predicts the existence of the WIMP alone without the rest of the SUSY model. This WIMP theory is motivated by solving the Boltzmann equation[9] for a particle with WIMP properties:

$$\frac{x}{Y_{eq}}\frac{dY}{dx} = -\frac{\eta_\chi^{eq}\langle\sigma\nu\rangle}{H(x)}[(\frac{Y}{Y_{eq}})^2 - 1] \tag{7}$$

This results in the same particle as the neutralino from the SUSY model.

$$\langle\sigma\nu\rangle_0 \sim 10^{-26} cm^3 s^{-1} \tag{8}$$

As gravitating particles, WIMPs are thought to cluster in spherical superstructures known as "halos" that encapsulate objects like galaxies and provide the gravitational corrections necessary to hold them together. Three paradigms exist to hopefully detect a WIMP: direct detection where the WIMP partakes in a particle process and is observed that way, indirect detection where gamma-rays thought to originate from DM annihilation would confirm its existence, or DM creation from a collision in a particle accelerator[10]. Since this report focuses on the LZ, it will investigate direct detection. The other two methods are beyond the scope of this review.

2.3 Direct Detection So Far

WIMPs in halos have low but non-zero velocities, which when combined with the relative velocity of the Earth through the galaxy cause a theorised background "DM wind" which the Earth is continuously traveling through. Direct detection of WIMPs involves detecting the impact of an incoming WIMP from this wind with a particle that it couples to, which is thought to produce heat and an emission photon and ionisation electron when the excited particle de-excites. Detectors can be designed for different coupling partners to explore different potential WIMP mass ranges. The choice of material for a given detector alters the precise techniques used to extract information from a DM collision. Owing to the rarity of predicted WIMP-SM collisions, direct detection experiments are held very deep underground to eliminate as much cosmic ray and surface radioactive background as possible, and are usually made of materials resistant to contamination by radioactive isotopes. The detection experiments up to the modern day have collectively constrained the possible mass and interaction parameters of the WIMP to a very narrow band, and this will no doubt continue as further experiments progress.[11]

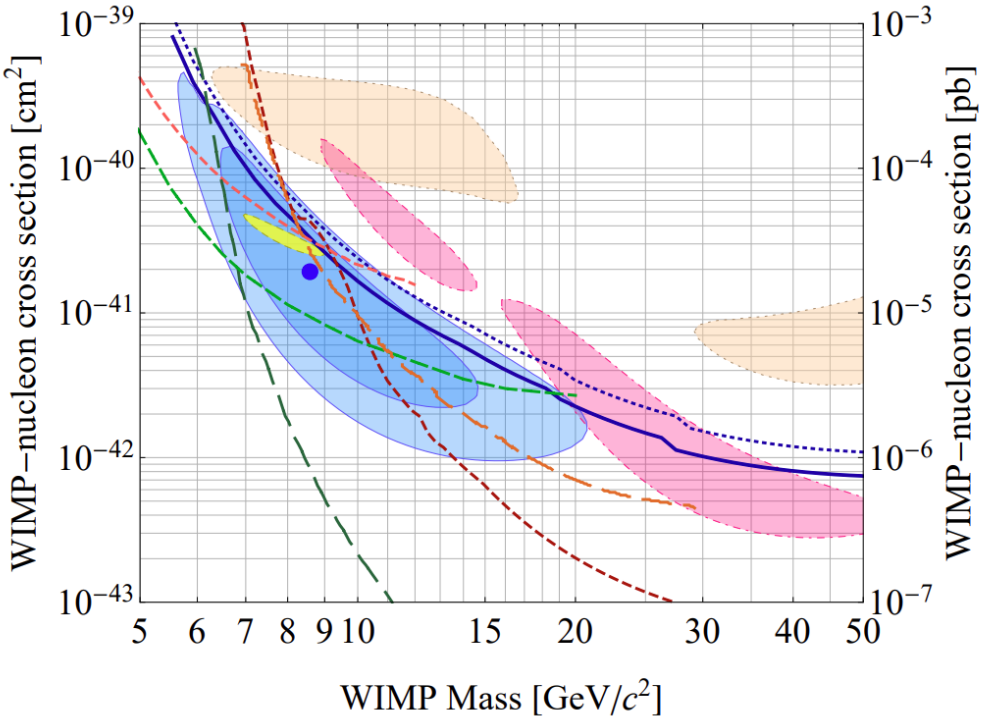


Figure 3: A plot of the WIMP mass-cross section relation as constrained by various detectors at 90% confidence level. Blue dotted line: CDMS II. Tan dotted line: DAMA-LIBRA.[12]

The DAMA/Nal and later DAMA/LIBRA experiments use thallium-doped sodium iodide scintillation crystals coupled to two PMTs each, which emit photons when subjected to ionizing radiation. The material is chosen for a combination of its strong scintillation, fast decay time, and cost-effectiveness.[13] Altering this setup is done by changing how the

crystals are grown. DAMA/LIBRA was conducted to attempt to detect DM through the annual modulation of the theorised DM signal as the Earth rotates around the Sun. The experiments have found a model-independent oscillating signal, but other experiments are in contradiction with its published results; therefore another upgrade to the experiment ("SABRE") is in progress to refine this matter, alongside two others, "COSINE-100" and "ANAIS-112". Thus far, none have managed to reproduce this result.[14][15][16]

The CoGeNT experiment used a single supercooled germanium crystal to reduce thermal noise while looking for WIMPs, which has merit as an apparatus that can double as an axion detector.[17] 56 days in, it detected a signal which could have been DM but was later found to not be replicable by alternative models of WIMP parameters.[18] Though it seemed to agree with DAMA's data, a later review showed that this signal could be explained by an underestimated background source.[19]

The CDMS and CDMS II experiments use a germanium and silicon cryogenic bolometer, respectively, to search for WIMPS. These bolometers read both a phonon and either a charge or scintillation to reconstruct the interaction; the CDMS II experiment set constraints according to Figure 3. A later iteration of the experiment using iZIP technology known as SuperCDMS proposed a limit of 1.4×10^{-44} cm^2 for a WIMP mass of 46 GeV/c^2 for its first run, with a second run underway.[20] The technologies were built starting with existing neutrino detectors[21] and achieve a combination of sensitivity to most possible WIMP masses and high signal purity due to the Ge-Si combination[22], making them useful as both a historical preliminary constraint[23] and to enact a search for low-mass neutrinos (<30 GeV).[20]

ArDM was a liquid noble gas detector, much like the LZ detector which is the focus of this review, but using argon as the detecting medium. The ArDM detector was lined with PMTs to detect scintillation and ionization events from an elastic collision between a liquid argon atom and a WIMP. Argon detectors work similarly to xenon detectors, but have an advantage in pulse shape discrimination between signal and noise[24] at the cost of having a naturally occurring isotope (^{39}Ar) which contributes a significant radioactive background to such experiments.[25] ArDM ran for three years as an attempt to develop a more robust argon detection technology[26], which is now used in the DEAP3600 and Darkside experiments currently in progress following the decommissioning of the ArDM project in 2019.[27]

3 The LUX-ZEPLIN Experiment

LUX-ZEPLIN is a DM direct detection experiment formed through the combination of the LUX and ZEPLIN research groups, which were separate liquid xenon WIMP experiments. LUX-ZEPLIN is an experiment which searches for WIMPs in the mass range of a few GeV up to many TeV, and has so far constrained the GeV WIMP mass candidate range to below $\frac{3\text{GeV}}{c^2}$.[28]

3.1 Detector Structure

At the heart of the LZ detector is the time projection chamber (TPC) which powers the analysis steps post-impact. The rest of the detector supports this central component. The detector uses a 7.62x6.1 m^2 water tank as the detector body to receive xenon. The outer detector is made of ten ultraviolet transmitting acrylic vessels filled with GdLS solution, which detect neutron-scattering events and allow them to be tagged as such in the TPC to reduce background noise. This background originates from the very low background radiation of the LZ's own components and the background radiation from the walls of the Davis Campus underground facility where the detector is located. Within the outer detector, two layers of cryostat made of low-radioactivity titanium are present to shield the TPC from unwanted neutron signals, with a thin layer of liquid xenon (LXe) acting as a filter for unwanted gamma signals. The acrylic vessels and liquid xenon cryostat together reduce the amount of NR background signals from 12.31 to 1.24 counts.[29]

In order to avoid introducing contamination from outside world elements that might mix with the detector components, they are first assembled outside of the detector in a clean lab room and then brought underground, either monolithically or in large but separate pieces that fit together without requiring underground welding, shielded by materials such as several layers of plastic bags to prevent contamination by dust and background on the way to the detector. Repairs are handled by carefully lifting the malfunctioning component out of the detector and transporting it back up to a clean lab, where it is repaired and then reinstalled according to this procedure.

3.2 Time Projection Chamber

The time projection chamber is a 7 tonne LXe WIMP target approximately 1.5 meters in diameter and height, which is viewed by two arrays of PMTs, all of which is contained in the inner cryostat vessel (ICV). The surrounding 2 tonnes of LXe in between the inner and outer cryostat vessels (OCV) are required to induce a dielectric gap between the contents of the ICV and the PMT array. This dielectric gap provides an additional veto against background neutrons and gammas, an effect known as "self-shielding" in detector literature. This gap also acts to prevent general photon leakage into the TPC from the general surroundings. A voltage of -50 kV is presently applied between the bottom and top of the TPC, with a goal of eventually getting this to -100 kV which would result in a drift field of 300V/cm within the TPC. To cool down the LXe, a cold trap of liquid nitrogen is used, achieving LXe temperatures around 173 K.

DM detectors usually contain scintillation, ionization, and heat emission in the same interaction, but can only properly focus on two out of three of these parameters with current technology. LZ focuses on the first two, leaving heat detection to other experiments. In practice, an ideal detection event would manifest as an interaction between a WIMP and a LXe atom, exciting either the atom or an orbiting electron and causing the release of a de-excitation photon and an ionization electron. The singular scintillation photon would be detected by the PMTs. The ionization electron would then drift towards the top of the TPC, causing a photon shower by Bremsstrahlung, also detected by the PMTs. By

applying reconstruction, a full 3D recreation of the detection event and its properties could be made.[30]

4 Enhancing Reconstruction Via Machine Learning

4.1 Data Interpretation and Reconstruction

In order to get any usable data out of a detection event in LZ, the detected signal must be reconstructed from PMT signals. PMTs work via the photoelectric effect, meaning they are subject to the rules of quantum mechanics. The emitted electron on photon impact is rebounded through layers of dynodes to cause gain in the signal, but as a quantum effect, there is a chance that emitted electrons do not reach the end of the PMT, potentially causing losses and leading to a statistic known as "quantum efficiency". Additionally, electrons travelling down the PMTs may impact dynodes at unfavourable angles. Due to these combining errors, the final result is not a delta-like peak, but rather a smooth near-Gaussian distribution which introduces uncertainty into the final measurement.[31]

The initial scintillation photon causes a singular peak in the output, and is known as the "S1" signal. The photon cascade from the drifting electron causes multiple peaks in the output which are always temporally after the S1 signal, known as the "S2" signal. By interpreting the combined S1 and S2 signals for a given detection event through parameters such as time between S1 and S2 signals and peak intensity, the position and nature of the detection event can be derived from the end data, which is the process known as "reconstruction".

Right now, existing reconstruction algorithms are very good with high NR signal purities sitting at 98%, but they run into a few issues. Firstly, the S1 and S2 signals can occasionally occur so closely to each other that they can overlap, which blurs data together. Secondly, multiple sources of data degradation such as electronic noise or varying PMT performance between detection events can further damage the sharpness of the peaks in the reconstructed data. The method to turn reconstruction events into a coherent data set involves a particularly capricious integral, so with current methods, analysis of LZ events must be done in bulk instead of individually, which further introduces data loss. This data is what is then plotted into a graph of S1 against S2 and binned to finally determine purity, showing that these forms of data loss do carry forward into results.[32]

4.2 Machine Learning

Machine learning (ML) is the use of algorithms that are able to process large supplied data sets to form a generalised algorithm that can be applied to unseen problems. A ML model starts off with an array of data sets with a known correlation or consistent pattern that the software architect wants the model to identify on which it is trained, then the model is progressively exposed to more and more known and fully understood data sets, where its performance is evaluated at regular intervals and its programming adjusted as needed. A ML model takes something such as a matrix as input and passes it into a sequence of "hidden layers", which are steps of the ML algorithm running one of several possible

"activation functions" that can have their precise operational steps adjusted by adjusting numerical values known as "parameters", after which the model produces an output which is evaluated against the target output for parameter adjustment.[33]

ML models are trained by adjusting parameters at each hidden layer, which may contain multiple nodes which act on the same input in different ways each, much like the connected neurons of a human brain. These parameters typically adjust the "weight" of the corresponding data that passes through each node, which transforms the magnitude to which the given data point will be able to influence the other data points as it is passed down the hidden layers, allowing the ML model to emphasise certain correlations that the algorithm discovers in the supplied data while de-emphasising unimportant effects.

A balance must be struck to avoid "overtraining" the ML model: giving it too many free parameters and too specific instructions will make the model exceed at analysing a given data set at the cost of no longer being able to analyse other unseen data sets.

To some degree, ML models train themselves, with human intervention limited to designing and parameterizing activation functions; past a certain point the inner workings of the ML model are unknown, which is why training it well is so important. A well-trained ML model is able to automate the analysis of huge data sets in which a correlation is sought after.[34]

4.3 Applying ML To Reconstruction

4.3.1 How Does This Help Reconstruction?

Given the aforementioned properties of ML models, it should come as no surprise that they can greatly improve reconstruction algorithms in detector physics. Detector data sets usually involve massive amounts of data affected by a lot of stochastic data loss in which an underlying pattern is sought out; this is the ideal scenario for which to apply a ML model. In particular for LZ, a ML model is able to process singular data points at incredible speeds, potentially allowing a paradigm shift from bulk evaluation of LZ data sets to counting each individual photon and then obtaining a net data set from these counted photons which would entirely eliminate the data loss associated with bulk evaluation.

ML models have already been applied to particle physics reconstruction with good result, and multiple models already exist with the specific goal of completing a classification or pattern recognition task, which is what LZ reconstruction can be boiled down to in trying to reverse engineer the position and type of impacting particle from output data. The simplest type of ML model for classification and pattern recognition is the "single-layer perceptron" (SLP), which takes an input vector of data points and returns a singular output after being ran through one activation function. An improved version of this model is the "multi-layer perceptron" (MLP), which is like the SLP except with the introduction of an arbitrary amount of hidden layers in between the input and output layers. In theory, a simple MLP could work for the task at hand, however other better ML algorithms have been developed using the MLP as a basis to implement "deep learning", which is a ML model that contains many, many hidden layers. One such model discussed and trialled in the cited paper is the RawNet ML model[35], a seventeen-step model which takes an input

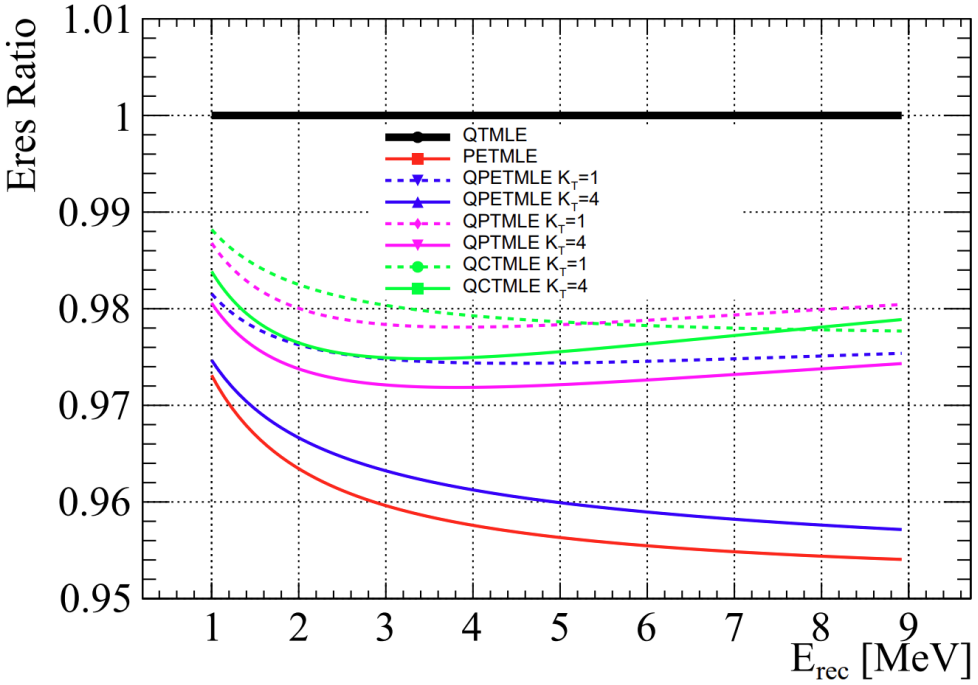


Figure 4: A table of outcomes for various implementations of the modified RawNet model, factoring in charge (Q), time (T), the model's PMT information (P), the model's confusion matrix (C), and photon events (PE) respectively. The test case of a prior iteration of the RawNet model is shown above in black, while various combinations of factors accounted for by the ML model are displayed in colour, showing a better energy resolution percentage than the test case in all cases.

waveform, extracts raw features from it then applies predicted adjustments to it, before applying three separate methods of denoising.

4.3.2 The RawNet Model

RawNet was trialled on Monte Carlo simulations of neutrino detections using software from the JUNO neutrino experiment, with the goal of discriminating and reconstructing the hit time of each separate photon event in the simulations. Though RawNet was designed as a vocoder, similarities exist between vocoding an audio sample and reconstructing a particle impact in a detector. This overlap combined with its prior performance in the particle physics role points towards potential application in the LZ detector to overhaul the photon counting algorithms.[36]

A paper has since been published applying a modified RawNet model to a simulated JUNO-like data set with the explicit aim of producing a model that reduces charge smearing and attempts to directly count the amount of photon events (PE) from waveforms; though the paper states that this particular model is strongest for low PE counts, it nonetheless improves the PE detection and mitigates some charge smearing on the simulated dataset, providing substantial evidence to deem RawNet a viable starting point for more specialised

ML photon counting models.[37]

4.3.3 The LCAG Transformer

Other ML techniques may have merit for DM detection; a 2022 paper describes the use of a transformer, which is a type of ML model that turns a semantic representation of a logical sequence such as a sentence or mathematical equation into a set of numerical "tokens" on which vector operations are performed to extract information, after which the data is finally converted back into semantic representation. The paper argues that particle decays, where an incoming particle causes a decay chain which is measured through the effects of the final child particles, constitute a form of logic tree and therefore can be represented in such a way that a transformer could operate on a detected event, through use of a new mathematical structure dubbed the "Lowest Common Ancestor Generations" (LCAG) matrix which is a representation of the ratio of particle decay steps (e.g. parent, first decay, second decay...) in the overall decay chain.

The paper operates on simulated Belle II and LHC-like data using one such transformer, which is shown to be able to classify decays of unseen data sets for trees of up to 10 leaves in size with approximately 47% confidence; this shows that this approach currently exhibits overtraining, but remains a sound proof-of-concept for future detector technologies to iterate on. Though dark matter detections usually do not involve noisy decay trees like those found in the LHC, the paper nonetheless displays a proof-of-concept of a classification ML model for particle physics, which could prove valuable to the DM detection effort.[38]

5 Conclusion

The true identity of DM remains a mystery, and despite efforts to detect the particle, no direct detection has been achieved as of today. As constraints get more and more precise, the window which can be searched for DM narrows ever further, and a marked upgrade in the technology powering the search for DM which can sidestep the intrinsic uncertainty in even the best detectors from stochastic effects and human limitations may become necessary to make tangible progress. The application of ML models such as the multi-layer perceptron seem suited to the classification tasks involved in particle data reconstruction; in particular, specialised algorithms such as RawNet and LCAG have shown promise when applied to tasks like particle physics event reconstruction, and being able to adapt such a technology to LZ might provide the sort of upgrades required to broaden the scope and refine the certainty with which LZ can probe the universe for WIMPs; failing that, they could provide a template for future experiments to start strong on new searches for new mass bounds of particle DM.

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